

Question 1

$$a) \frac{x-2}{x^2-x-6} = \frac{3}{2x+4} - \frac{x}{4-x^2}$$

$$\frac{x-2}{(x-3)(x+2)} \checkmark = \frac{3}{2(x+2)} \checkmark + \frac{x}{(x-2)(x+2)} \checkmark \quad x \neq 2; x \neq -2; x \neq 3 \checkmark$$

$$\text{LCD} = 2(x+2)(x-2)(x-3) \checkmark$$

$$2(x-2)(x-2) = 3(x-2)(x-3) + 2x(x-3) \checkmark$$

$$2(x^2-4x+4) = 3(x^2-5x+6) + 2x^2-6x$$

$$2x^2-8x+8 = 3x^2-15x+18 + 2x^2-6x \checkmark$$

$$2x^2-3x^2-2x^2-8x+15x+6x+8-18=0 \checkmark$$

$$-3x^2+13x-10=0$$

$$3x^2-13x+10=0 \checkmark$$

$$(3x-10)(x-1)=0$$

$$\underline{x = \frac{10}{3}} \checkmark$$

or

$$\underline{x = 1} \checkmark$$

(10)

$$b) 1 - \frac{4}{\sqrt{x+2}} = \frac{32}{x+2}$$

$$\text{let } k = \sqrt{x+2} \checkmark$$

$$\text{then } k^2 = x+2$$

$$\sqrt{x+2} = -4$$

N/A

$$1 - \frac{4}{k} = \frac{32}{k^2} \checkmark$$

$$k^2 - 4k - 32 = 0 \checkmark$$

$$(k+4)(k-8) = 0$$

$$\underline{k = -4} \checkmark$$

$$\text{or } \underline{k = 8} \checkmark$$

$$\sqrt{x+2} = 8$$

$$x+2 = 64$$

$$\underline{x = 62} \checkmark$$

(6)

$$c) x^2 - x = 14 - \frac{24}{x(x-1)}$$

$$x^2 - x = 14 - \frac{24}{x^2 - x}$$

$$\text{let } h = x^2 - x \quad \checkmark$$

$$h = 14 - \frac{24}{h}$$

$$h^2 = 14h - 24$$

$$h^2 - 14h + 24 = 0 \quad \checkmark$$

$$(h - 2)(h - 12) = 0$$

$$\therefore h = -2 \quad \checkmark$$

$$\text{or } h = 12 \quad \checkmark$$

(8)

$$\therefore x^2 - x = -2$$

$$x^2 - x + 2 = 0$$

$$(x + 1)(x - 2) = 0 \quad \checkmark$$

$$\underline{x = -1} \quad \text{or } \underline{x = 2}$$

$$x^2 - x = 12$$

$$x^2 - x - 12 = 0$$

$$(x + 3)(x - 4) = 0$$

$$\underline{x = -3} \quad \text{or } \underline{x = 4}$$

$$d) x^2 + 9 = 0$$

$$x^2 = -9 \quad \checkmark$$

$$x = \pm \sqrt{-9} \quad \checkmark \checkmark$$

$$= \pm 3i$$

(2) (3)

27 marks

Question 2

a) $\frac{x^2+4}{x-2} < 0$ ↗ always ↘ positive

$\therefore x-2 < 0$ ✓
 $\therefore x < 2$ ✓ (3)

b) $\frac{3-x}{x^2+9x-10} \geq 0$

$\frac{x-3}{(x-1)(x+10)} \leq 0$ ✓

OR number line.

	-10	1	3	
$x-3$	-	-	-	0 + +
$x-1$	-	-	-	u/D + + +
$x+10$	-	u/D	+	+
	-	u/D	+	+
	-	u/D	+	+

$\therefore x < -10$ ✓ on $1 < x \leq 3$ ✓

(10)

Question 3

a) i) $|x+1| = -2$

No soln

✓ (1)

ii) $|2x+1| = 4$

$\therefore 2x+1 = 4$

$2x = 3$

$x = \frac{3}{2}$ ✓

or $2x+1 = -4$

$2x = -5$

$x = -\frac{5}{2}$ ✓

✓ (2)

iii) $|x-3| + 2x = -2$

$|x-3| = -2 - 2x$

if $x-3 \geq 0$
 $x \geq 3$ ✓

then

$x-3 = -2-2x$

$3x = 1$ ✓

$x = \frac{1}{3}$ N/A

if $x-3 < 0$

i.e. $x < 3$ ✓

then

$-x+3 = -2-2x$

$x = -5$ ✓

(4)

$\therefore x = -5$

b) i) $y = a|x-p| + q$ ✓

$y = a|x-0| + 0$ ✓

Sub in (3; 2) ✓

$2 = a|3|$

$\therefore a = \frac{2}{3}$

$\therefore$

$y = \frac{2}{3}|x|$ ✓

(3)

$$ii) y = a|x-p| + q$$

$$y = a|x-1| - 1 \quad \checkmark$$

sub in $(0, -4)$ $\checkmark$

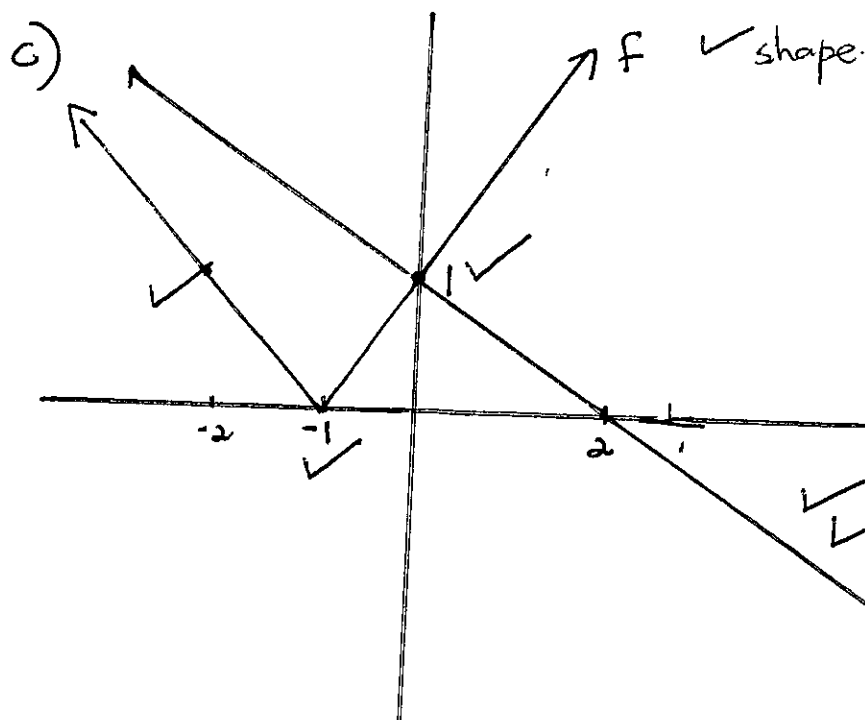
$$-4 = a|0-1| - 1$$

$$-3 = a|1|$$

$$a = -3$$

$$\therefore y = -3|x-1| - 1$$

$\checkmark$ (3)



(6)

$$x+1 = -\frac{x}{2} + 1$$

$$2x+2 = -x+2$$

$$3x = 0$$

$$x = 0$$

$$\text{or } -x+1 = -\frac{x}{2} + 1$$

$$-x + \frac{x}{2} = 0$$

$$-2x+x = 0$$

$$-x = 0$$

$$x = 0$$

From graph

$$iii) f(x) < g(x)$$

absolute value graph below str line.

$$-4 < x < 0$$

(2)

Question 4

$$\begin{aligned} \text{a) i) } (2+3i)(4-5i) &= 8 + 2i - 15i^2 \checkmark \\ &= 8 + 2i + 15 \checkmark \\ &= 23 + 2i \checkmark \quad (3) \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{(5+5i)}{(2-i)} \times \frac{(2+i)}{(2+i)} &\checkmark \\ &= \frac{10 + 15i + 5i^2}{4 - i^2} \checkmark \\ &= \frac{10 + 15i - 5}{4 - (-1)} \\ &= \frac{5 + 15i}{5} \checkmark \\ &= 1 + 3i \checkmark \quad (4) \end{aligned}$$

● b) $x = 2 - i$
then $x = 2 + i$

$\therefore x - 2 + i$ is a factor. $\checkmark$
so is $x - 2 - i$ $\checkmark$

$$\begin{aligned} (x-2+i)(x-2-i) &\text{ is a factor } \checkmark \\ x^2 - 2x - ix - 2x + 4 + 2i + ix - 2i + 1 &\checkmark \\ &= x^2 - 4x + 5 \end{aligned}$$

● $\therefore (x^2 - 4x + 5)(2x - 3) = 2x^3 + ax^2 + bx - 15$

$$\begin{aligned} \therefore -8x^2 - 3x^2 &= ax^2 \\ 12x + 10x &= 22x \end{aligned}$$

$$\begin{aligned} \therefore a &= -11 \checkmark \\ \therefore b &= 22 \checkmark \end{aligned}$$

(8)

or do using sum and product to find quadratic.

c) i) $x = 2 - 3i$

$x = 2 + 3i$

$\therefore \text{sum} = 4$ ✓✓

prod = $(2-3i)(2+3i)$
 $= 4 + 9$
 $= 13$ ✓✓

or any other method.

$\therefore$ quad eqn is $x^2 - 4x + 13 = 0$. ✓✓

ii) $(x^2 - 4x + 13)(x - 5) = x^3 - 9x^2 + 33x - 65$ ✓✓

$\therefore \underline{x = 5}$ or $x = \underline{2 \pm 3i}$ (4)

25 marks

Question 5

a) $g(x+2) = 3(x+2) + 5$ ✓
 $= 3x + 6 + 5$
 $= \underline{3x + 11}$ ✓

b) $h^{-1}(x) = ?$

$h(x) = g(g(x))$
 $= g(3x+5) + 5$
 $= 3(3x+5) + 5$
 $y = 9x + 15$ ✓✓

$f(-12) = |-12| = 12$ ✓✓

$\therefore 3x + 11 = 12$

$3x = 1$

$x = \frac{1}{3}$ ✓ (5)

c) $x + f(x) = 0$

$x + |x| = 0$

$|x| = -x$ ✓✓

if $x \geq 0$

then $x = -x \therefore 2x = 0$

$\underline{x = 0}$ ✓

if $x < 0$

then $-x = -x \therefore x \in \mathbb{R}$

but $x < 0$

$\therefore x \leq 0$

✓

(5)

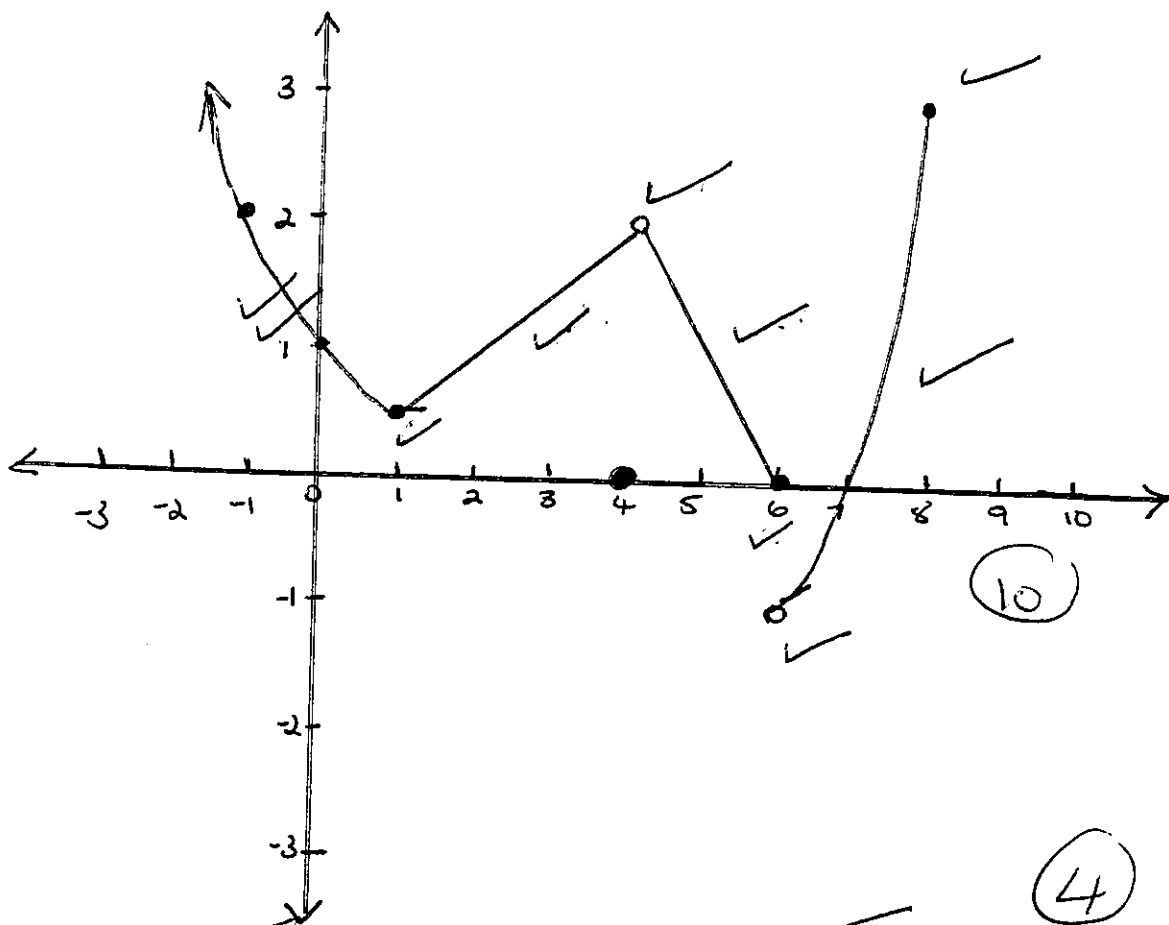
16 marks

$h^{-1}(x) = x = 9y + 15$
 $x - 15 = 9y$
 $y = \frac{1}{9}x - \frac{15}{9}$
 $y = \frac{1}{9}x - \frac{5}{3}$ ✓✓

$\therefore \underline{h^{-1}(x) = \frac{1}{9}x - \frac{5}{3}}$ (6)

QUESTION 6

a)



b) $f(1) = \frac{1}{2}$; $f(2) = 1$; $f(4) = 0$; $f(5) = 1$

c) $x = 6$ or $x = 7$ or $x = 4$

d) $f(f(6)) = f(0) = 2^{-0} = 1$

e) $x \in (-\infty; 8]$ Range: $y \in (-1; \infty)$

f) $x = 4 \rightarrow$ Removable.
 $x = 6 \rightarrow$ Jump.

Question 67

a) $120^\circ \times \frac{\pi}{180^\circ} = \frac{2\pi}{3} \checkmark \checkmark$ (2)

b) $\frac{7\pi}{6} \times \frac{180^\circ}{\pi} = 210^\circ \checkmark \checkmark$ (2)

c) $\sec^2 x - \sec x - 2 = 0$
 $(\sec x + 1)(\sec x - 2) = 0.$

$\sec x = -1 \checkmark$ or $\sec x - 2 = 0 \checkmark$
 $\cos x = -1 \checkmark$

$x = \pi + k, 2\pi \checkmark$

$\sec x = 2$
 $\cos x = \frac{1}{2} \checkmark$ (8)

$x = \frac{\pi}{3} + k, 2\pi \checkmark$

$x = -\frac{\pi}{3} + k, 2\pi \checkmark$

d) LHS = $\frac{1}{\tan x \sqrt{1 - \cos^2 x}}$
 $= \frac{1}{\frac{\sin x}{\cos x} \sqrt{\sin^2 x}} \checkmark$
 $= \frac{1}{\frac{\sin^2 x}{\cos x}} \checkmark$
 $= \frac{\cos x}{\sin^2 x} \checkmark$

RHS = $\operatorname{cosec} x \cdot \cot x$
 $= \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} \checkmark$
 $= \frac{\cos x}{\sin^2 x} \checkmark$

(8)

$\therefore \underline{\underline{\text{LHS} = \text{RHS}}}$

$$e) A = \frac{1}{2} r^2 \theta$$

$$72 = \frac{1}{2} r^2 \theta$$

$$\theta = \frac{72}{\frac{1}{2} r^2}$$

$$S = r\theta$$

$$36 = r\theta$$

$$\theta = \frac{36}{r}$$

$$\therefore \frac{72}{\frac{1}{2} r^2} = \frac{36}{r}$$

$$72r = 36 \times \frac{1}{2} r^2$$

$$72r = 18r^2$$

$$0 = 18r^2 - 72r$$

$$0 = 18r(r - 4)$$

$$\therefore r = 0$$

N/A

$$\text{or } r = 4$$

129

Question 18

$$a) \lim_{x \rightarrow 4} \frac{x^2 - 16}{4 - x}$$

$$= \lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{(4-x)}$$

$$= -8$$

4

$$b) \lim_{x \rightarrow \infty} \frac{5 - 3x - 2x^2}{(2x - 3)^2}$$

$$= \lim_{x \rightarrow \infty} \frac{5 - 3x - 2x^2}{4x^2 - 12x + 9}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{5}{x^2} - \frac{3}{x} - 2}{4 - \frac{12}{x} + \frac{9}{x^2}}$$

6

$$c) f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(x+h)^2 + 3(x+h) - 1 - (2x^2 + 3x - 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(x^2 + 2xh + h^2) + 3x + 3h - 1 - 2x^2 - 3x + 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + 3h}{h}$$

$$= \lim_{h \rightarrow 0} (4x + 2h + 3) = 4x + 3$$

6

16 marks